MILITARY SPECIFICATION

INDICATOR, PHASE SEQUENCE, SWITCHBOARD AND PANEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers phase sequence indicators which indicate on polyphase alternating current (a.c.) circuits the order in which the voltage wave in each phase reaches its maximum positive value.

1.2 Classification. Phase sequence indicators shall be of the following types as specified (see 6.2):

MR	4S	W	4	AC	PHS
Component	Style	Color scheme	Normal voltage	Frequency	Unit
1.2.1	1.2.2	1.2.3	1.2.4	1.2.5	1.2.6

1.2.1 Component. Indicators covered by this specification are designated by the two-letter symbol "MR".

1.2.2 Style. The style is designated by a two-digit number which signifies the size and shape as follows.

1.2.2.1 First digit.

2 - 2-1/2 inch indicator with two indicator lamps
3 - 3-1/2 inch indicator with five indicator lamps
4 - 4-1/2 inch indicator with five indicator lamps

1.2.2.2 Second digit.

R - Round flange
S - Rectangular flange

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Color scheme.

B - Black dial, white lettering
W - White dial, black lettering

1.2.4 Nominal voltage.

1 - 120 volts (V)
2 - 208 V
4 - 450 V

1.2.5 Frequency.

AC - 60 hertz (Hz)
AF - 400 Hz
AM - 25 to 1000 Hz

1.2.6 Unit. The unit shall be PHS for phase sequence.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-80 - Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-G-3787 - Glass, Laminated, Flat (Except Aircraft).
MIL-M-16034 - Meters, Electrical-Indicating, (Switchboard and Portable Types).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-129 - Marking for Shipment and Storage.

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STANDARDS (Continued)

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-810 - Environmental Test Methods.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Copies of specifications, standards and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C39.1 - Requirements for Electrical Analog Indicating Instruments.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection (see 4.3 and 6.3).

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of the best quality used in commercial practice and shall be free from all defects and imperfections.

3.2.1 Restricted materials. Materials restricted from Navy use for the construction of phase sequence indicators shall be as specified in MIL-E-917.

3.2.2 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.3 Safety. The safety requirements shall be in accordance with MIL-E-917.

3.4 Construction.

3.4.1 Case dimensions and mounting. The case for 4-1/2 inch indicators shall conform to the outline and mounting dimensions shown on figure 1. Other cases shall conform to ANSI C39.1.

3.4.2 Case material. The case shall be constructed of metal or plastic. The case material shall be in accordance with MIL-M-16034.

3.4.3 Painting. Parts not fabricated of a corrosion-resisting material or having a corrosion-resistant treatment as specified in MIL-E-917, except for semi-conductors, shall be painted as specified in MIL-E-917. Only one coat of gray enamel shall be applied on the outside of the enclosure. Touching-up will be permitted for marks or scratches due to factory handling.

3.5 Operating voltages. The indicators shall operate over the voltage range of plus or minus 7 percent of nominal voltage (see 6.2). Applied voltage of 140 percent shall not damage the indicator.

3.6 Frequency. The indicators shall operate on either a 60 or 400 Hz supply, as specified (see 6.2). Indicators which operate on range of frequencies from 25 to 1000 Hz are acceptable.

3.7 Terminals. Three terminals shall be provided for rear connection and marked "A", "B" and "C".

3.8 Indicating function. Indicators shall contain two or five indicator lights conforming to requirement 50 of MIL-STD-454. The light indicating "A-B-C" phase sequence shall be labeled "Correct Ø Sequence", and the other light indicating "C-B-A" phase sequence, "Incorrect Ø Sequence". The remaining three lights, when required shall have a clear or white lens and shall be arranged so that all three are lighted only when all three indicator terminals are energized. These three lights shall be labeled "All Lamps On When All Phases Connected". The indicator front shall be similar in appearance to that shown on figure 2. Background color coding shall be as shown on figure 2.

3.9 Performance characteristics. The performance characteristics shall be as specified herein.

3.9.1 Reference temperature. The reference temperature shall be 25 degrees Celsius (°C).

3.9.2 Dielectric strength. The indicators shall withstand the dielectric strength test of 4.6.2 without damage or flashover.

3.9.3 Finite resistance to ground. When tested in accordance with 4.6.3, indicators shall operate correctly as specified in 3.8.

3.9.4 Insulation resistance. When tested in accordance with 4.6.4, the insulation resistance of the indicators shall be not less than 20 megohms.

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3.9.5 Humidity. Indicators shall operate satisfactorily following exposure to the humidity test of 4.6.5.

3.9.6 Vibration. When tested in accordance with 4.6.6, the indicators shall operate satisfactorily and shall show no evidence of breakage, permanent deformation or loosening of parts.

3.9.7 Shock. When tested in accordance with 4.6.7, the indicators shall show no evidence of breakage or loosening of parts.

3.9.8 Endurance. After all other tests have been performed, the equipment shall be subjected to the endurance test of 4.6.8. Upon completion of the endurance test, the indicators shall function as specified in 4.6.1.

3.10 Identification and marking. The indicators shall be permanently and legibly marked on the dial or front surface. The following information is required (see figure 2):

(a) Manufacturer's name and trademark, or both.
(b) Manufacturer's type or model number, or both.
(c) Federal stock number.
(d) Indicator light labeling as specified in 3.8.

3.11 Reliability. The reliability of phase sequence indicators shall be evaluated by a part stress analysis prediction in accordance with MIL-HDBK-217.

3.12 Workmanship. Parts shall be properly and securely assembled. Surfaces of the case shall be smooth and free from flash projections and roughness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performing all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performing the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall consist of three indicators of the type covered by this specification, and shall be subjected to the examination of 4.5 and the tests specified in 4.6.

4.4 Quality conformance inspection. Each of the indicators shall be subjected to the examination of 4.5 and the test specified in 4.6.1 and examined for quality of workmanship as specified in 3.12. Any indicator which contains one or more defects shall not be offered for delivery.

4.5 Examination. The indicator shall be examined to determine conformance with the following:

(a) Material (see 3.2).
(b) Saftey (see 3.3).
(c) Design, construction, dimensions and mounting (see 3.4 and figure 1).
(d) Labeling (see 3.8 and 3.10).
(e) Workmanship (see 3.12).

4.6 Test procedures.

4.6.1 Operation. Phase sequence indicators shall be connected to a three-phase a.c. power source to verify compliance with 3.5, 3.6 and 3.8. Tests shall be performed at normal voltage and plus or minus 7 percent of normal voltage after applying 140 percent voltage for 2 minutes.

4.6.1.1 Connection of two phases. Indicators shall be checked with only two leads or terminals energized to verify that the requirements for indication of incomplete connections specified in 3.8 are met.

4.6.2 Dielectric strength. The indicators shall be subjected to a test voltage of 1900 volts root mean square applied between all external live parts of the indicator and (a) the case, if metallic, or the metallic mounting panel if the case is nonmetallic, and (b) all exposed metal parts not live parts of the indicator. The voltage shall be a nominal 60 Hz in frequency and shall approximate as closely as possible a true sine wave in form. The voltage shall be applied continuously for 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage directly from the circuit or indirectly by means of a voltmeter coil on the transformer or an auxiliary meter transformer. To avoid voltage surges, the applied voltage shall be gradually raised to its full value and gradually reduced upon completion of the 60-second test.

4.6.3 Grounded phase. Indicators shall be checked for correct operation when:

(a) Resistors of 500 ohms are connected between each indicator terminal and the indicator case mounting bolts. (This test determines correct operation on a distribution system with finite ground resistance.)
(b) Each of the three resistors specified in (a) are shorted in turn. (This test determines correct operation on a distribution system with a fault.)

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4.6.4 Insulation resistance. With 1000 volts direct current, the insulation resistance shall be determined between all external live parts and (a) the case, if metallic, or the metallic mounting panel if the case is nonmetallic and (b) all exposed metal parts not live parts of the indicator.

4.6.5 Humidity resistance. Indicators shall be tested in accordance with MIL-STD-810, method 507, procedure IV for humidity resistance.

4.6.6 Vibration resistance. Indicators shall be tested in accordance with the type I requirements of MIL-STD-167-1 for vibration resistance.

4.6.7 Shock. The indicators shall be tested in accordance with the grade B, class I, type A requirements of MIL-S-901 for shock.

4.6.8 Endurance test. After the testing of 4.6.1 through 4.6.6 has been completed, the indicators shall be subjected to the following endurance test: each of the sample indicators shall be connected to a test circuit that, for each indicator, cycles from proper to improper phase sequence at the rate of 6 cycles per hour for a total of 40 hours.

4.7 Inspection of packaging. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packaging. Indicators, and mounting hardware if required, shall be individually cushioned and unit-packaged in accordance with method IIb of MIL-P-116, without the use of contact preservatives. The packaging shall conform to level A or C of MIL-E-17555 as specified (see 6.2).

5.2 Packing. Packing shall conform to level A or C of MIL-E-17555, as specified (see 6.2).

5.2.1 Domestic shipment-immediate use. The packing for domestic shipment-immediate use indicators shall conform to level C of MIL-E-17555.

5.2.2 Domestic shipment and storage or overseas shipment. Indicators packaged for domestic shipment and storage or for overseas shipment shall conform to level A of MIL-E-17555.

5.3 Marking. In addition to any special marking required (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The indicators are for determining the phase sequence of three-phase a.c. power systems.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Classification (see 1.2).
(c) Sample for first article inspection required (see 3.1).
(d) Voltage (see 3.5).
(e) Frequency (see 3.6).
(f) Preservation and packaging, packing, and marking requirements if other than as specified in 5.1 (see 5.2).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 When a manufacturer can present proof of a previous approval of an identical indicator, first article inspection may be waived for the production of the next 5000 indicators or 5 years, whichever occurs first. After this event, the manufacturer will be required to submit a sample for first article inspection.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

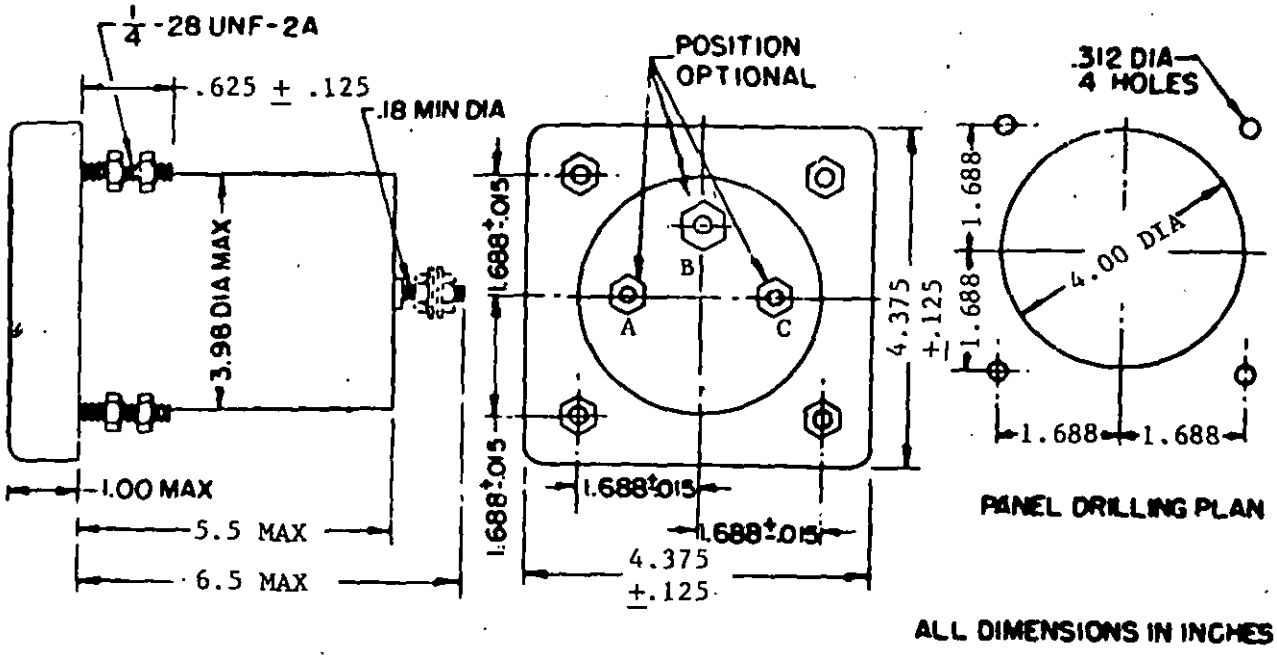
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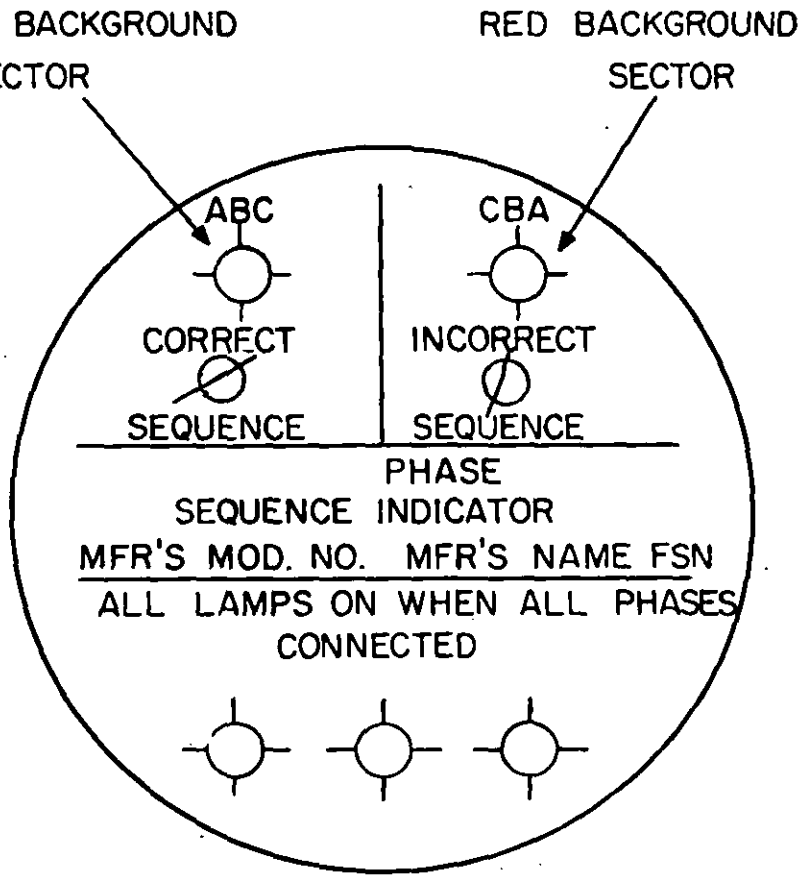
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SH 4943

FIGURE 1. Design, construction, and physical dimensions of phase sequence indicator.

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SH 9542

Note: 2.5-inch indicators only require the correct and incorrect sequence lights. 3.5- and 4.5-inch indicators shall contain all five lights arranged in a similar manner as shown above.

FIGURE 2. Phase sequence indicator.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-16103F	INDICATOR, PHASE SEQUENCE, SWITCHBOARD AND PANEL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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